End-to-End Data Management Pipeline for a Recommendation System

RecoMart Recommendation System

Team: Group 49

Team Member Details:

1. K Sharnika - 2025AB05310

2. Akshay Govindrao Telang - 2025DA04003

3. Debosmita Bose - 2025AB05316

4. Sachin Sharma - 2025DA04006

5. Marati Hareesh Babu - 2025AB05317

Submission Date: 17.07.2026

1. Problem Statement

RecoMart wants to create a scalable recommendation engine that uses user-item interactions and product metadata to provide personalized product suggestions and improve customer engagement and cross-selling opportunities.

2. Objectives

- Support batch ingestion from MovieLens files and near-real-time product ingestion from a REST API into timestamp-partitioned raw storage

- Validate schema, missing values, duplicates, ranges, and timestamp formats

- Prepare data and generate rating-distribution, popularity, sparsity, and correlation analysis

- Engineer recommendation features and store them in CSV and SQLite

- Maintain versioned feature retrieval and DVC-based data lineage

- Train and evaluate an NMF recommendation model and expose a command-line inference interface

- Orchestrate the pipeline with Airflow and track experiments with MLflow

3. Methodology / Pipeline

The implemented workflow supports daily batch ingestion and five-minute near-real-time REST API ingestion, followed by validation -> preparation and EDA -> feature engineering and SQLite storage -> versioned feature store -> NMF model training and evaluation -> inference. DVC records stage dependencies and output versions, Airflow orchestrates both ingestion modes, and MLflow records model parameters, metrics, and artifacts.

4. Implementation Details

- Ingestion scripts process the MovieLens 100k public dataset for batch interactions and product metadata, fetch supplemental product data from a REST API with retries, and write timestamp-partitioned batch data under data/raw/api and independent micro-batches under data/raw/near_real_time_api with separate manifests.

- Validation checks verify required schemas, missing values, duplicate rows, rating ranges, timestamps, prices, and popularity values, then generate JSON and PDF data-quality reports.

- Preparation joins product metadata with interactions, handles missing values, encodes categories, safely standardizes numeric attributes, and generates the required EDA plots.

- Feature engineering creates user activity frequency, user and item average ratings, and item popularity features, and stores transformed feature tables in SQLite using database/schema.sql.

- The custom feature store records feature definitions, sources, transformations, available versions, and versioned retrieval paths for training and inference.

- DVC tracks the pipeline stages, dependencies, generated outputs, and reproducible lineage through dvc.yaml and dvc.lock.

- The NMF model uses a per-user positive-interaction holdout, excludes previously rated products from recommendations, and reports RMSE, Precision@5, Recall@5, and NDCG@5.

- Airflow provides the daily recomart_end_to_end DAG and the five-minute recomart_near_real_time_ingestion DAG with validation and failure logs, while MLflow stores the experiment run ID, parameters, metrics, model metadata, model artifact, and feature table.

5. Results and Outputs

Metric	Value
Model	NMF
NMF Components	3
RMSE	3.4376
Precision@5	0.0338
Recall@5	0.1688
NDCG@5	0.1201

Evaluation method: per-user positive interaction holdout; training rows = 99058; test rows = 942.

Validation summary: 2 datasets checked, passed = True

Dataset	Rows	Missing	Duplicates	Issues
interactions	100000	0	0	0
products	1682	0	0	0

Latest batch ingestion partition: N/A; files recorded = 3.

Latest near-real-time REST API ingestion: 2026-07-17T18:50:09.086286+00:00; files recorded = 1.

Feature store version: v20260717185250; available versions = 1; rows = 100000; documented features = 10.

Structured feature storage:

SQLite Table	Rows
user_features	943

SQLite Table	Rows
item_features	1682
interaction_features	100000

Experiment Tracking

MLflow run id: bdd55505e803454daf732903b8eb6b2a

Local MLflow database:
C:\Users\hareesh\Desktop\DMML-Assignment-main\RecoMart-Recommendation-System\mlflow.db

Local artifact path:
C:\Users\hareesh\Desktop\DMML-Assignment-main\RecoMart-Recommendation-System\mlruns

Sample Recommendations (top 5 per user):

- 196: 50, 100, 258, 127, 181

- 186: 50, 181, 1, 313, 7

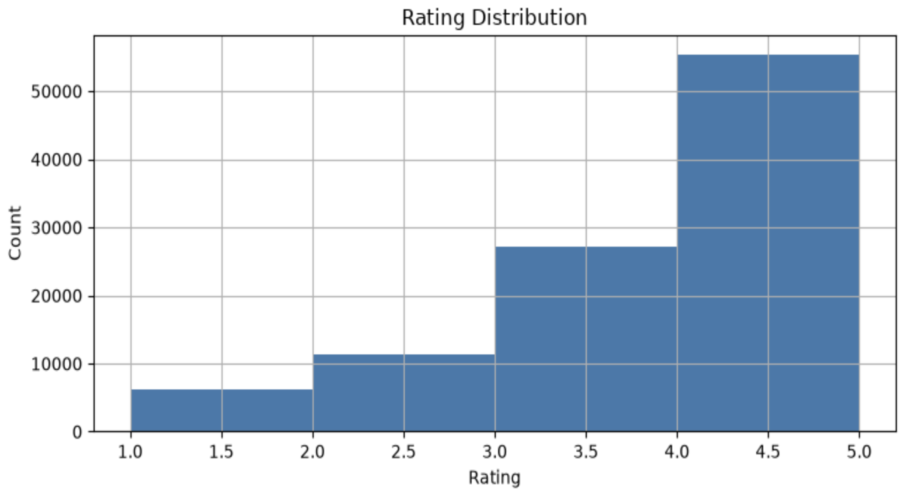
- 22: 82, 22, 69, 56, 98

- 244: 98, 79, 210, 96, 195

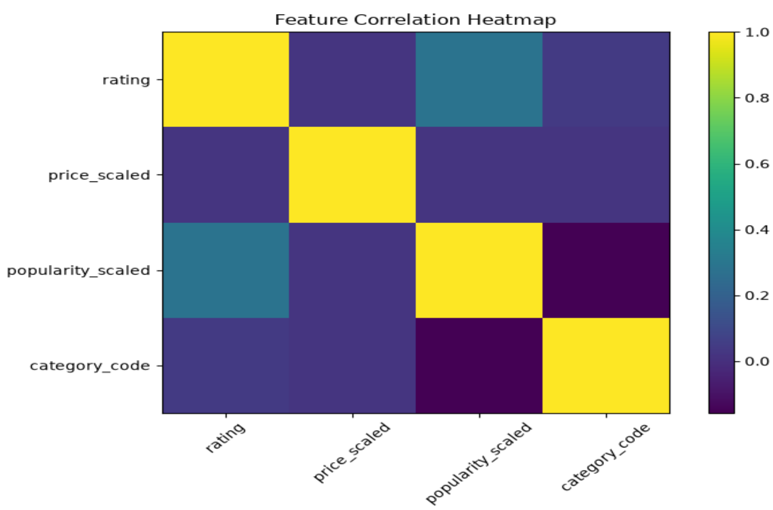
- 166: 100, 313, 50, 237, 302

6. EDA Plots and Execution Evidence

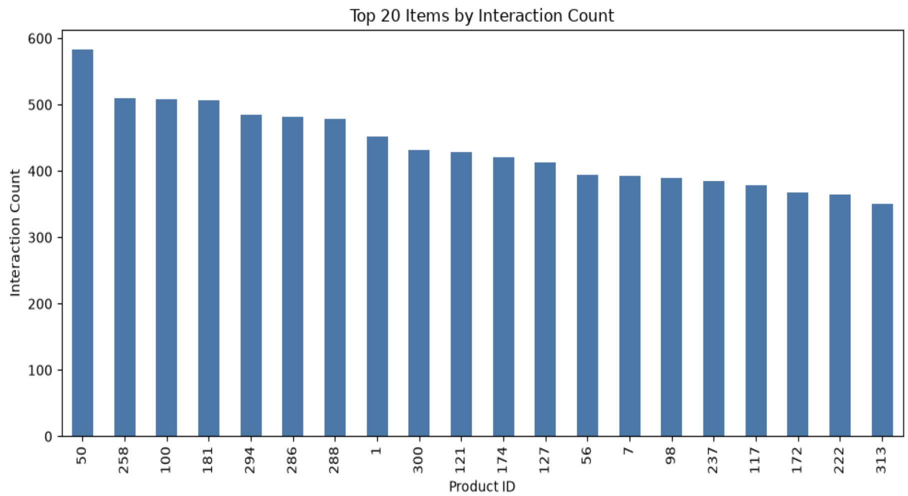
Rating distribution



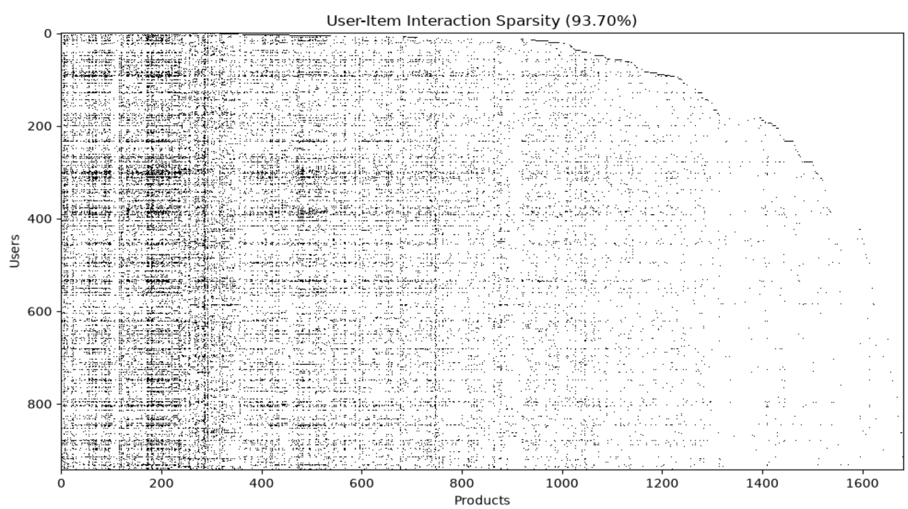
Feature correlation



Item popularity



Interaction sparsity



Airflow, MLflow, and DVC screenshots were not available when this report was generated. Add airflow_success.png, mlflow_run.png, and dvc_dag.png under reports/screenshots before final submission.

7. Conclusion and Future Scope

The pipeline provides a modular and reproducible foundation for recommendation-system data management, including batch and near-real-time REST API ingestion, validation, EDA, structured feature storage, feature versioning, holdout evaluation, inference, orchestration, and experiment tracking. Future work can add Kafka-based event streaming, cloud object storage, richer hybrid recommendation features, automated model promotion, and deployment through a production inference API.

8. Submission Links

Google Drive Link to Video Walkthrough: Not provided

Google Drive Link to Deliverables ZIP: Not provided

Repository: https://github.com/hareeshm24/DMML-Assignment